

MATH I (Fall 2026)

WEEK 2: INDEPENDENT STUDY ASSIGNMENTS (Homework)

About the Independent Study Assignments

The Independent Study Assignments are homework problems and assignments that will help you learn the week's content more deeply by solving problems and expanding your knowledge and skills. *Your answers will not be submitted and will not be graded.* You may solve the problems individually or collaborate with your fellow students or ask the assistance from an AI tool (to help you clarify the concepts, not providing the solution without you thinking about it first).

Week 2 - Assignments

The following problems/projects cover the material presented in Week 2. Choose to do as many as you wish (the more the better):

Assignment 1. [Optional assignment]

The following is an *optional assignment* – *you can engage with it if you are familiar with AI-assistants, or if you want to start learning what AI-assistants can do for you in a context-specific domain.* The exercise is as follows:

Interact with an AI-assistant (such as ChatGPT, Claude, DeepSeek, Gemini, etc.), by structured conversation, through the following steps:

a) Think of the most important point of the lecture's content and ask the AI-assistant to evaluate your answer (you may ask the AI-assistant to evaluate it (usually, they do the evaluation in terms of content and clarity, but you can also ask for depth and complexity, or other criteria you may choose). The AI-assistant will not grade your input but it will provide content and perspectives to help you better understand the material.

The following example of a simple prompt shows how you can ask (prompt) the AI-assistant:

Your (simple) prompt: "In the second lecture of Math I course, in which Chapters 1.1-1.5 of the OpenStax Calculus 1 textbook were covered, the most important points are ...[PLEASE, COMPLETE this, with what YOU CONSIDERED as the most important concepts/methods covered]." Can you evaluate this statement of mine?

b) Create your own problem (on the lecture's material) and ask the AI-assistant (of your choice) to evaluate it in terms of difficulty and depth.

c) Solve your proposed problem by yourself. You may discuss your problem with fellow students. d) Ask the AI-assistant for a solution to your problem but check its validity (that is, check if the AI-assistant's solution is correct).

If you find wrong outcomes, discrepancies or contradictory statements, converse with the AI-assistant to mention these points, asking for explanations, to resolve them. If you cannot resolve the wrong outcomes, please consult the instructor.

e) Compare your solution with AI-assistant's solution.

Assignment 2: Relations and functions

Determine whether each relation defines y as a function of x . Explain your answer.

a. $y = 3x - 2$, b. $x^2 + y^2 = 9$,

c. $y^2 = x + 1$, d. $x = y^3 - 2$.

Where appropriate, solve for y and apply the vertical-line test.

Assignment 3: Function notation

Let

$$f(x) = 2x^2 - 3x + 1.$$

Calculate and simplify:

$$f(0), \quad f(-2), \quad f(a), \quad f(a+h),$$
$$f(a+h) - f(a), \quad \frac{f(a+h) - f(a)}{h}, \quad h \neq 0.$$

The final expression anticipates the difference quotient used in differential calculus.

Assignment 4: Domain

Find the natural domain of each real-valued function:

a. $f(x) = \frac{x+1}{x^2-9}$,

b. $g(x) = \sqrt{5-2x}$,

c. $h(x) = \frac{\sqrt{x-1}}{x-4}$,

d. $p(x) = \sqrt{\frac{x+2}{x-1}}$.

Express each domain using interval notation.

Assignment 5: Range and graph

Find the range of each function:

$$f(x) = x^2 + 2, \quad g(x) = -|x - 1| + 4, \quad h(x) = \sqrt{x + 3} - 2.$$

Assignment 6: Algebra of functions

Let

$$f(x) = x^2 - 1, \quad g(x) = \sqrt{x + 2}.$$

Determine:

$$f + g, \quad f - g, \quad fg, \quad \frac{f}{g}.$$

For each new function, state its domain.

Assignment 7: Composite functions

Let

$$f(x) = 2x - 3, \quad g(x) = x^2 + 1.$$

Find:

$$(f \circ g)(x), \quad (g \circ f)(x), \quad (f \circ f)(x), \quad (g \circ g)(x).$$

Are $f \circ g$ and $g \circ f$ equal? Explain why function composition is generally not commutative.

Assignment 8: Symmetry

Determine whether each function is even, odd or neither:

$$x^4 - 2x^2, \quad x^3 - 4x, \quad x^3 + x^2, \\ |x| + 1, \quad \frac{x}{x^2 + 1}.$$

Verify your conclusions algebraically and graphically.

Assignment 9: Identify the function class

Classify each function as linear, polynomial, rational, algebraic or transcendental. A function may belong to more than one category.

$$3x - 7, \quad x^4 - 2x + 1, \quad \frac{x + 1}{x^2 + 4}, \\ \sqrt{x^2 + 1}, \quad 2^x, \quad \sin x.$$

Assignment 10: Linear engineering model

The temperature inside an industrial chamber increases linearly from 20°C at $t = 0$ to 80°C at $t = 15$ minutes.

1. Find a function $T(t)$ describing the temperature.

2. Interpret its slope and vertical intercept.
 3. Find $T(8)$.
 4. At what time does the chamber reach 60°C ?
 5. State a physically reasonable domain.
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Assignment 11:

Study the examples (marked as Independent Study Assignments) in lecture presentation (see Week 2 lecture's pdf file)

Assignment 12: Polynomial behaviour

For

$$p(x) = x^3 - 4x$$

and

$$r(x) = \frac{x+1}{x-2},$$

determine:

- domain;
- intercepts;
- symmetry, if any;
- long-run or end behaviour;
- locations where the graph is undefined.

Use graphing software to check your analytical results.

Assignment 13: Piecewise electricity tariff

An electricity provider charges according to

$$C(E) = \begin{cases} 0.10E, & 0 \leq E \leq 200, \\ 20 + 0.15(E - 200), & E > 200, \end{cases}$$

where E is monthly energy consumption in kWh and $C(E)$ is the cost.

1. Find $C(150)$, $C(200)$ and $C(350)$.
 2. Sketch the graph.
 3. Is the cost model continuous at $E = 200$?
 4. Explain the meaning of each slope.
 5. Suggest one limitation of the model.
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Assignment 14: Degrees and radians

Convert:

$$30^\circ, \quad 135^\circ, \quad 300^\circ$$

to radians, and convert

$$\frac{\pi}{6}, \quad \frac{5\pi}{4}, \quad \frac{7\pi}{3}$$

to degrees.

Assignment 15: One-to-one functions

Determine whether each function is one-to-one on its stated domain:

$$f(x) = 3x - 5, \quad x \in \mathbb{R};$$

$$g(x) = x^2, \quad x \in \mathbb{R};$$

$$h(x) = x^2, \quad x \geq 0;$$

$$p(x) = \frac{1}{x}, \quad x \neq 0.$$

Use the horizontal-line test where appropriate.

Assignment 16: Finding inverse functions

Find the inverse of each function and state the domains of the function and its inverse:

$$f(x) = 4x - 7,$$

$$g(x) = \frac{x+2}{3},$$

$$h(x) = x^2 + 1, \quad x \geq 0,$$

$$p(x) = \frac{2x-1}{x+3}, \quad x \neq -3.$$

Verify each answer using composition:

$$f(f^{-1}(x)) = x.$$

Assignment 17: Exponential growth and decay

For each model, decide whether it represents growth or decay and identify its initial value and growth or decay factor:

$$A(t) = 500(1.08)^t,$$

$$B(t) = 1200(0.85)^t,$$

$$C(t) = 40e^{0.03t},$$

$$D(t) = 200e^{-0.4t}.$$

Assignment 18: Solve exponential equations

Solve:

$$2^{x+1} = 16,$$

$$5^{2x} = 125,$$

$$e^{3x} = 7,$$

$$100e^{-0.2t} = 20.$$

Give exact logarithmic answers where necessary, followed by decimal approximations.

Assignment 19: Battery discharge

The stored energy in a battery is modeled by

$$E(t) = E_0 e^{-0.025t},$$

where t is measured in hours.

1. What fraction of the initial energy remains after 10 hours?
 2. When will 50% of the initial energy remain?
 3. When will 10% remain?
 4. Explain why the model never predicts exactly zero energy.
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Assignment 20: Population growth

A bacterial population is modeled by

$$P(t) = 800e^{0.35t},$$

where t is measured in hours.

1. Find the initial population.
2. Find the population after 5 hours.
3. Determine the doubling time.
4. Explain whether the model can remain realistic indefinitely.

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